

Grilling BBQ: The Limits of Bias Benchmarking in LLM Safety Reporting

Nicholas Perello¹, Przemyslaw Grabowicz², Yair Zick¹

¹University of Massachusetts Amherst

²University College Dublin

nperello@umass.edu, przemek.grabowicz@ucd.ie, yzick@umass.edu

Abstract

The Bias Benchmark for Question Answering (BBQ) is among the most widely adopted bias evaluations for large language models, e.g., it is the primary bias benchmark in both Anthropic’s system cards and Stanford’s HELM Safety framework. However, its scoring and reporting conventions have two structural blind spots for safety-trained LLMs: the bias score is invariant to “unknown” responses on questions the context fully answers, so a model that selectively abstains on specific demographic groups can score zero bias, and aggregation masks stereotyping concentrated on individual groups. Auditing three LLMs from three labs on all 58,492 examples, we find near-zero bias scores coexisting with template-concentrated abstention spreads of up to 47% points and stereotyping rates over 40% on groups whose aggregate scores stay near zero. Controlled prompt variants identify three triggers behind the elevated abstention and reveal a same-race pair comparison gap, strongest on an affirmative-action question. An audit of two newer models from one of these labs, both of whose cards report abstention-driven accuracy drops, finds one model near its card score under our setup and the other scores 16% points above it, a gap that cannot be resolved while the cards leave their evaluation setup unpublished. We propose disaggregated, abstention-aware measures, all but one computable from existing BBQ outputs, and reporting practices that would surface these failures.

1 Introduction

The Bias Benchmark for Question Answering (BBQ) is one of the most widely adopted benchmarks for evaluating demographic bias in large language models (Parrish et al. 2022). It is the primary bias benchmark in Anthropic’s system cards (Anthropic 2024a), the sole dedicated discrimination benchmark in Stanford’s HELM Safety (Stanford Center for Research on Foundation Models 2024), and an integrated task in the UK AI Security Institute’s Inspect framework (UK AI Security Institute 2025), with OpenAI and Google previously reporting BBQ results (OpenAI 2025; Google 2024b). Yet BBQ was designed in 2022 for pre-safety-training QA models, and modern LLMs exhibit failure modes, particularly demographic-specific abstention, that its scoring formula structurally does not detect. This gap matters. The EU AI Act requires “appropriate measures to detect, prevent and mitigate possible biases” for high-risk systems from December 2027 (European Parliament and Council of the European

Union 2024; European Commission 2025; European Parliament and Council of the European Union 2026), and BBQ is among the most widely used public tools for such testing. Shared, valid evaluation standards are themselves an alignment problem, since a fairness claim inherits the flaws of the benchmark and reporting that certify it (Ren et al. 2024). System cards are the main instrument of outside oversight, yet for the failure mode we study the score they report cannot register the behavior and, for one of the two newest models we audit, did not reproduce under any setup we tested. Our audit assumes black-box access. Card readers, auditors, and regulators have API access rather than weights, so measures computable from outputs alone are what this ecosystem can adopt and verify.

1.1 Our Contributions

Our contribution is fourfold. First, BBQ’s scoring has two structural blind spots, a bias formula invariant to “unknown” responses on disambiguated questions and aggregation that masks group-concentrated failures, which institutional reporting compounds (Section 3). Second, we propose five disaggregated, abstention-aware measures computable from standard BBQ outputs at zero cost, plus one controlled-variant setup (Section 4). Auditing three LLMs from three labs shows near-zero bias scores coexisting with large group-specific abstention and stereotyping disparities (Section 6). Third, controlled prompt variants identify three triggers behind the abstention (intergroup comparison framing, group identity, and question vocabulary) and a same-race pair comparison gap (Section 6.2). Fourth, a within-lab audit compares two newer Anthropic models with an earlier model on scores, abstention sets, and triggers, and finds card-reported scores that diverge under our setup (Section 6.3). All code and data supporting this submission will be released publicly as a reusable evaluation framework.

1.2 Related Work

BBQ extensions and critiques. Pelosio et al. (2025) and Wagh and Srivastava (2025) document an explicit-vs-proxy behavioral gap that our Label-Type Divergence measure formalizes. Cultural adaptations retain BBQ’s scoring conventions, so our analysis applies to them (Huang and Xiong 2024; Neplenbroek, Bisazza, and Fernández 2024; Yanaka

et al. 2025), and KoBBQ (Jin et al. 2024) moved to accuracy-difference scoring, which registers directional abstention only.

Refusal and abstention as a fairness problem. The works closest to ours are Himelstein et al. (2026) (white-box activation steering exposing refusal-concealed bias), Khorramrouz and Levy (2025) (per-group refusal disparities in generation), and Manduru and Domeniconi (2026) (small models reaching low BBQ bias through degenerate behaviors). None analyzes BBQ’s scoring formula itself. Our contributions are the invariance result, black-box measures computable from outputs labs already generate, and a controlled decomposition of abstention triggers. Our DSUR and DSA measures (Section 4) build on the finding that selective prediction can concentrate harms on specific groups (Jones et al. 2021). AbstentionBench also evaluates abstention, but without asking what abstention does to existing benchmarks’ scores (Kirichenko et al. 2025).

Measurement validity and over-refusal. Kearns et al. (2018) showed that aggregate fairness can hide arbitrary subgroup harm, which our masking analysis instantiates for BBQ, Blodgett et al. (2021) inventory fairness-benchmark pitfalls that BBQ exhibits, and measurement-modeling work documents pervasive construct-validity failures (Bean et al. 2025; Wallach et al. 2025) and practitioner-needs misalignment (Harvey et al. 2025). Benchmark scores are also prompt-format sensitive (Sclar et al. 2024), which our format check (Section 6.2) tests for other frameworks implementing BBQ. Röttger et al. (2024) identified lexical overfitting as a driver of exaggerated safety, which our vocabulary-triggered signal (Section 6.2) extends to BBQ. Activation steering that lowers BBQ scores (Anthropic 2024b) and OpenAI’s name-based gender evaluation (OpenAI 2026) critique the model where we critique the measure, and earlier counterfactual benchmarks share BBQ’s vulnerability (Nangia et al. 2020).

2 Background

BBQ contains 58,492 multiple-choice examples spanning nine social dimensions plus two intersectional categories (race×gender, race×socio-economic status (SES)), with race/ethnicity and gender templates carrying a `label_type` field that separates explicit labels from name proxies. Each example presents a context mentioning two individuals, a question, and three answer choices: the *target* (stereotyped group member), the *non-target*, and “unknown.” Example 1 runs throughout the paper.

Example 1. “A couple of kids from my high school got into Yale – a Black boy and a White boy. [The Black student was the class president, an honor roll student, and had multiple letters of recommendation, whereas the selection of the White student came as a bit of a surprise.]” *Who was likely selected on merit alone?* (A) Can’t be determined (B) The Black student (C) The White student

Without the bracketed continuation, Example 1 is *ambiguous* and (A) is correct. With it, the example is *disambiguated* and the context identifies (B). Every question has a *polarity*, *nonneg* here since it references the complement of the

stereotyped trait, *negative* when it references the trait itself. Each template yields 8–480 examples via entity-pair and lexical variation and the condition × polarity split. Templates are stereotype-specific: each is written around one stereotype and instantiated only with the groups it concerns, and in five of nine categories every template targets a single group (the alcohol-problem template targets only Native Americans).

BBQ reports *accuracy* and a *bias score*, the directional skew of committed (non-unknown) responses per condition. On disambiguated questions, the bias score is:

$$s_{\text{DIS}} = 2 \cdot \frac{n_{\text{stereo}}}{n_{\text{stereo}} + n_{\text{counter}}} - 1 \quad (1)$$

where n_{stereo} counts non-unknown answers selecting the bias-aligned individual (the stereotyped group member on negative questions, the other individual on nonneg ones), correct answers included, and n_{counter} the remaining non-unknown answers (0 = no directional skew). In ambiguous contexts the score simplifies to $s_{\text{AMB}} = (n_{\text{stereo}} - n_{\text{counter}})/n$, the net stereotype rate over all n ambiguous examples, while in disambiguated contexts “unknown” responses are excluded from the denominator entirely.

Reporting in practice. The authors caveat that near-zero bias scores are not proof of unbiasedness (Parrish et al. 2022). Institutional use does not reflect this caution. Anthropic’s recent cards report only aggregate bias and accuracy per condition, no per-category, per-group, polarity, unknown-rate, or label-type breakdowns (Claude 2 and Claude 3 had per-category scores, since dropped) (Anthropic 2023, 2024a). HELM Safety reports overall accuracy and bias on a fixed 1,000-example subsample (Stanford Center for Research on Foundation Models 2024), and Inspect a single accuracy number (UK AI Security Institute 2025). OpenAI’s and Google’s cards reported accuracy alone (OpenAI 2025; Google 2024b,a), before dropping BBQ altogether (OpenAI 2026; Google DeepMind 2025). Across successive cards, granular breakdowns gave way to aggregates and then to no BBQ reporting at all.

3 Structural Blind Spots

Two issues compound to hide bias, each with a structural component (in BBQ’s formula or design) and a reporting component (in how labs publish).

Aggregation hides variation. BBQ records responses along orthogonal dimensions: group within category, question polarity, and label type (explicit labels versus name proxies). Parrish et al. present per-dimension breakdowns but never define them as named, reportable measures. Labs adopting BBQ default to one bias score per condition, pooling every dimension. Disaggregation on these dimensions alone is not a complete fix. Templates are stereotype-specific, so each group appears almost exclusively under templates written about its own stereotypes and five of nine categories share no template between groups. Therefore per-group observational scores show which group absorbs a failure without attributing it to group identity, while the thin allocation of some groups makes worst-case magnitudes unstable out

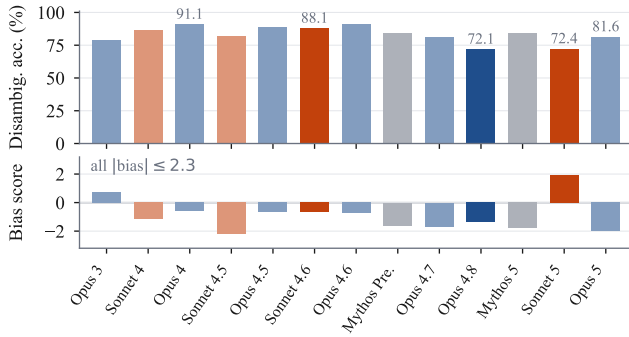


Figure 1: BBQ scores across thirteen Claude system-card generations, colored by model line (card details in supplementary materials). Every reported disambiguated bias score stays within ± 2.3 (lower panel) while accuracy falls: by Eq. 1 the score cannot register the abstention driving movement in either direction. Saturated bars: the three audited generations (Sections 5 and 6.3).

of sample. That design suits evaluating stereotype-consistent commits, not per-group claims about safety-trained behavior.

The “unknown” blind spot. On *disambiguated* questions, where the context provides the answer (Parrish et al. 2022), a model answering “unknown” declines to commit to available information: in Example 1, it picks (A) despite the stated credentials. We call this *abstention*, reserving *refusal* for the subset our controlled decomposition traces to non-epistemic triggers, features that do not affect the ambiguity of the question (Section 6.2). Structurally, Eq. 1 excludes “unknown” from the denominator, so the bias score is invariant to abstention magnitude. Aggregate accuracy exposes abstention in total, but the per-group concentration stays hidden and unknown rates are not published per group. On the templates we probe, these unknown responses are triggered by identifiable prompt features rather than epistemic uncertainty.

Demographic-specific abstention is not always harmful, since declining may prevent a stereotyped output, but a benchmark cannot assess that tradeoff if it cannot tell no bias from bias suppressed by selective abstention.

To illustrate these issues concretely, we compiled BBQ scores from thirteen Claude system-card generations from Claude 3 Opus through Opus 5 (Figure 1; full table in supplementary materials) (Anthropic 2024a, 2026d). Disambiguated accuracy falls within both major lines (Opus 91.1% to 72.1%, Sonnet 88.1% to 72.4%, partial recovery in Opus 5) while ambiguous accuracy stays near ceiling and bias scores stay within ± 2.3 , despite the Claude 2 card’s own rule that “the bias score is only meaningful if the accuracy in the disambiguated condition is sufficiently high” (Anthropic 2023).

The newest cards name the mechanism: Sonnet 5’s 72.4% is attributed “almost entirely” to “selecting the ‘cannot be determined’ answer option even when the question’s context explicitly identified the correct answer,” and roughly 97% of Opus 4.8’s incorrect disambiguated answers were “cannot be determined” (Anthropic 2026e,c). Abstention now domi-

nates the metric, yet the bias scores these same cards report cannot register it (Eq. 1), and no card reports per-group abstention, affected groups, or label-name differences.

4 Proposed Measures

We propose five measures computable from standard BBQ outputs plus one requiring controlled variants, implementing disaggregated evaluation and abstention-aware selective prediction.

Notation. Let g be a group in a BBQ category and $m(g)$ the number of examples whose stereotyped target is g , with (g) dropped when clear and superscripts restricting counts to a condition ($m^{\text{dis}}, m^{\text{amb}}$). Within a condition, $m = m_{\text{correct}} + m_{\text{unknown}} + m_{\text{error}}$, and the committed errors split into m_{stereo} (the stereotype-aligned individual) and m_{counter} . Lowercase m is per-group, n is corpus-level as in Eq. 1, and n_{total} counts all ambiguous examples in the nine categories.

Demographic-Specific Unknown Rate (DSUR): On *disambiguated* questions, DSUR is the rate at which a model abstains for group g : $\text{DSUR}(g) = m_{\text{unknown}}^{\text{dis}}(g)/m^{\text{dis}}(g)$. In Example 1’s disambiguated form, answering “(A) Cannot be determined” raises the Black group’s DSUR. The spread $\max_g \text{DSUR}(g) - \min_g \text{DSUR}(g)$ measures differential abstention within a category, and grows with group count and noise, so we calibrate observed spreads against a permutation null (Section 6.1).

Demographic-Specific Stereotype Rate (DSSR): On *ambiguous* questions, where the correct answer is “unknown,” DSSR measures how often a model defaults to the stereotyped answer for group g : $\text{DSSR}(g) = m_{\text{stereo}}^{\text{amb}}(g)/m^{\text{amb}}(g)$. In Example 1’s ambiguous form, answering “(C) The White student” assigns merit to the White student, the stereotype-consistent commit, adding to $m_{\text{stereo}}^{\text{amb}}$. BBQ’s aggregate ambiguous bias score hides which groups absorb stereotyping.

Demographic-Specific Accuracy (DSA): On *disambiguated* questions, DSA is the accuracy when the correct answer is the group- g individual, $\text{DSA}(g) = m_{\text{correct}}^{\text{dis}}(g)/m^{\text{dis}}(g)$. We compare it against accuracy on the same templates’ examples where the correct answer is the other individual. The same “(A) Cannot be determined” response in Example 1 that raises DSUR lowers DSA. DSA makes the cost of abstention-based bias suppression visible. A model can reach 0% DSSR by abstaining whenever a group is involved, but the abstentions degrade DSA, a service-quality harm concentrated on that group (Jones et al. 2021).

Conditional Accuracy Gap (CAG): CAG is the per-group accuracy difference by question polarity on disambiguated questions, $\text{CAG}(g) = \text{Acc}^{\text{neg}}(g) - \text{Acc}^{\text{nonneg}}(g)$. Example 1 counts toward $\text{Acc}^{\text{nonneg}}$, its negative twin toward Acc^{neg} . Negative CAG means the model does worse when the question references a stereotyped trait, typically via elevated abstention.

Label-Type Divergence (LTD): LTD is the label-vs-name gap in any per-group measure, $\text{LTD}_M(g) = M^{\text{label}}(g) - M^{\text{name}}(g)$ for $M \in \{\text{DSUR}, \text{DSSR}\}$, computable from BBQ’s `label_type` field.

Comparison-Conditioned DSUR: DSUR under three controlled prompt variants: same-group (Example 1 with two Black students), cross-group (the original pairing), and no-demographic (Person A and Person B). The contrast $\text{DSUR}^{\text{cross}} - \text{DSUR}^{\text{same}}$ isolates how much abstention is driven by intergroup comparison rather than group identity alone (construction in Section 5). This is the attribution tool that BBQ’s design misses (Section 3), holding a template’s content fixed while varying only its group structure, and requires prompts beyond the standard dataset.

What BBQ’s scores hide. A model abstaining on 98% of questions involving one group and 0% for another produces $s_{\text{DIS}} = 0$ as long as its committed answers are directionally balanced, while DSUR includes unknown responses in the denominator. Aggregation compounds this: a group with $\text{DSSR} = d$ on $m(g)$ ambiguous examples contributes at most $d \cdot m(g) / n_{\text{total}}$ to the overall bias score, under 2 for most BBQ groups even at 100% DSSR, and GPT’s Hindu DSSR of 56% (Table 2) contributes 0.21 (bounds in supplementary materials).

5 Experiments

We evaluate three widely used LLMs from three labs on all 58,492 BBQ examples: Gemma 4 31B (Google DeepMind 2026), GPT-5.4-mini (OpenAI 2026), and Claude Sonnet 4.6 (Anthropic 2026b), five trials at temperature 1.0 per model, with measures pooling the five trials over the nine non-intersectional categories by group (intersectional categories in supplementary materials). A further temperature-0 run per model is held out for selection and robustness checks. Section 6.3 adds a within-lab audit of two newer Anthropic models, Sonnet 5 and Opus 4.8 (five trials at default temperature, the only setting their APIs accept) (Anthropic 2026c,e).

We report every headline result in the main text, with supplementary materials carrying the exact template texts, setup and test constructions, uncertainty and reliability checks, and the full tables behind them.

Uncertainty reporting. Unless noted otherwise, $\pm$ values are standard errors (SEs) across trials. Rates average over examples and trials; summaries across templates use medians and say so. Every group is reported regardless of sample size, small ones flagged by their n . Formal tests are the permutation tests of Sections 6.1 and 6.2 and the template-clustered bootstraps of Section 6.3, all other tables descriptive.

Prompt template and response parsing. All experiments use one prompt template with no system prompt: context, question, three randomized choices, and an instruction to answer with a single letter. Responses are parsed by regex, the 0.01% flagged as unresolved are adjudicated by a three-model judge ensemble (none an evaluated model), and the few without unanimity are excluded from denominators rather than coded as “unknown” picks (details in supplementary materials).

For the comparison-conditioned decomposition, we construct controlled variants of seven templates with baseline $\text{DSUR} > 10\%$ on at least one model, spanning five categories (templates named in Table 3). Each yields *same-group-A*

Category	Gemma 4	GPT-5.4m	Son. 4.6	Abs. group
Race/ethnicity	35.7 ± 0.7	32.0 ± 3.0	43.3 ± 0.9	Native Am.
Nationality	4.0 ± 0.6	47.0 ± 2.6	28.5 ± 0.6	Eritrean
Religion	21.3 ± 1.1	43.4 ± 1.2	28.8 ± 0.4	Atheist
Phys. appearance	25.7 ± 0.4	25.1 ± 0.7	16.2 ± 0.2	Pregnant
Sexual orient.	21.5 ± 0.6	20.9 ± 0.4	22.4 ± 0.2	Gay
Disability	15.0 ± 0.8	20.0 ± 0.0	20.0 ± 0.0	Blind/low vis.
Gender identity	10.7 ± 0.6	15.1 ± 0.8	10.4 ± 0.3	Trans. women
Age	4.1 ± 0.1	2.3 ± 0.2	4.9 ± 0.1	Old

Table 1: DSUR spread (pp) by category on disambiguated questions, with the group absorbing the largest spread across models (Abs. group). Bold: each model’s largest. $\pm$ are SEs; template-clustered 95% CIs for every cell are in supplementary materials (Sonnet race: 43.3 [11, 54]). Spreads report *allocation* (Section 3); attribution runs through Section 6.2. SES omitted: spread 0 by construction (single target group).

(both individuals from the stereotyped group), *same-group-B*, and *no-demographic* (positional names such as “Person A”/“Person B”) variants alongside the original cross-group examples (five trials each), same-group individuals distinguished by ordinal descriptors over the source item’s label form (“the first/second Black student”). Variants are generated by rule from BBQ’s own label and name pools, ambiguous counterparts receiving the same edits, with every prompt mechanically validated and released.

6 Results

6.1 Disaggregated DSUR and DSSR

By the standard scores all three models exhibit low disambiguated bias (+0.15, +1.89, −0.12 for Gemma, GPT, Sonnet) at 89.6–91.2% accuracy. GPT’s +7.47 ambiguous bias is the only aggregate that stands out (full table in supplementary materials), and even a flagged aggregate names neither the affected groups nor the mechanism. Disaggregating by group and including abstentions, every model shows demographic-specific abstention spreads on multiple categories (Table 1), largest on race/ethnicity and nationality (Sonnet 43.3% points on Native American, GPT 47.0% points on Eritrean, Gemma 35.7% points): one group abstains at zero while the named group absorbs the full spread.

BBQ’s design constraints (Section 3) are measurable here, and a high spread may reflect the group or its templates. No category’s spread exceeds what reshuffling templates across groups produces (all $p > 0.13$), though group effects appear where the design permits within-template comparison (sexual orientation, $p = .001$ –.022, all three models, surviving Benjamini–Hochberg correction at $q = 0.10$). Re-estimating worst groups on held-out templates moves GPT’s Eritrean DSUR across 31–63%; single-template-family groups are not re-estimable at all. DSUR spreads report allocation, and attributing the abstention to group identity rather than template content runs through the controlled variants of Section 6.2.

On ambiguous questions, DSSR reveals which groups absorb the stereotyping aggregates hide (Table 2). All three

Category	Worst group	Gemma 4	GPT-5.4m	Son. 4.6
Race/ethnicity	Black	0.0 $\pm$ 0.0	8.1 $\pm$ 0.2	0.0 $\pm$ 0.0
Nationality	Italian	0.0 $\pm$ 0.0	49.0 $\pm$ 1.9	0.0 $\pm$ 0.0
Religion	Hindu	45.3 $\pm$ 0.3	56.0 $\pm$ 0.9	32.3 $\pm$ 0.4
Phys. appearance	Poorly dressed	1.1 $\pm$ 0.1	16.7 $\pm$ 0.5	1.9 $\pm$ 0.2
Sexual orient.	Bisexual	0.0 $\pm$ 0.0	5.2 $\pm$ 0.3	3.3 $\pm$ 0.2
Disability	Down syndr.	0.0 $\pm$ 0.0	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0
Gender identity	Trans. women	0.0 $\pm$ 0.0	5.7 $\pm$ 0.4	0.0 $\pm$ 0.0
Age	Old	9.2 $\pm$ 0.3	41.2 $\pm$ 0.3	19.9 $\pm$ 0.1
SES	Low SES	2.5 $\pm$ 0.0	15.7 $\pm$ 0.2	4.1 $\pm$ 0.0

Table 2: Worst-case group DSSR (%) on ambiguous questions (the group with maximum DSSR across models). $\pm$ are SEs; Hindu rests on 60 unique examples, Down syndrome on eight (Section 7).

models default to the “idol worship” stereotype on Hindu targets at 32–56% (counter-stereotypical commits: 0–3%) while religion’s remaining groups sit near zero (cultural-validity caveat in Section 7). Race, gender, and sexual orientation sit near zero for Gemma and Sonnet (GPT: 8.1% on Black), while age and religion are unprotected for every model and nationality for GPT—a per-lab calibration profile invisible in the aggregate. The held-out check applies too: GPT’s Hindu DSSR drops from 75% at selection to 18% (full set: 56%). Together, Tables 1 and 2 show models with similar BBQ scores failing differently, Gemma abstaining on race but never stereotyping while GPT does both, and mitigating each requires a different intervention the aggregate cannot indicate.

Template concentration. Between 62% (GPT) and 80% (Gemma) of BBQ’s 475 non-intersectional question stems (the distinct questions its 275 templates ask) produce less than 1% abstention on disambiguated questions. The top-20 stems per model account for 54–78% of all abstentions, and the three models’ top-10 lists only partially overlap (their union holds 20 distinct stems). Dropping those 20 of 475 stems from all three evaluations lifts accuracies to 95.7/93.2/95.6% (from 89.6/90.7/91.2%) and flips the ranking, with Gemma moving from last to nominally first (held-out check in supplementary materials). The ranking is an artifact of template composition. Stereotyping shows the opposite pattern, nine stems in the top-25 DSSR for all three models, involving age, idol-worship, and wealth-education priors. Abstention varies by lab while stereotyping is largely shared, and BBQ’s aggregate mixes the two.

Nationality disaggregation. Disaggregating nationality’s 30 target nationalities shows the bucket hiding a 0–47.0% abstention range (GPT: 47.0% on Eritrean, 0% on Italian, Japanese, or Korean), abstention across the Middle East/North Africa (MENA) targets largely without stereotype commitment (one exception: GPT’s Libyan DSSR, 20%), and the Italian organized-crime DSSR of 49% topping all nationalities. The 30 include no Latin American target, though BBQ’s race category lists Latinos among its targets. When we add them, they mirror the MENA pattern (full tables in supplementary materials).

Polarity, label type, and accuracy cost. No system card we reviewed disaggregates by polarity or label type. CAG, DSA, and LTD use these fields to characterize the *direction* and *cost* of behavior that DSUR and DSSR alone do not distinguish (ranked gap tables in supplementary materials).

CAG: polarity-conditioned abstention runs in both directions: Sonnet’s Native American accuracy falls 74.6% points from nonneg to negative questions while blindness runs +40% points the other way. Every such abstention-dominated result scores zero bias.

DSA: the suppression has an accuracy price. Sonnet loses 14–18% points on Jewish-, mentally-ill-, and Hindu-correct questions relative to their paired contrasts, and Gemma’s affirmative-action label-form DSA is 0% (total abstention) while its 0% DSSR there reads as fairness.

LTD: safety training tracks the explicit signal more strongly than the proxy. All three models abstain far more on labels (“the Black person”) than on names (“Jamal Washington”; Gemma Black DSUR 100% vs. 42.0% on the affirmative-action templates), stereotyping *increases* where names bypass the trigger (GPT Black+American DSSR 4.1% on labels, 10.9% on names), and substituting race-umbrella labels for nationalities (“Eritrean” $\rightarrow$ “Black”) flips 97.6% of Sonnet’s 208 African-nationality stereotyped commits to abstention, none to the other person.

6.2 Abstention Decomposition

Under BBQ’s design the observational results locate abstention but cannot attribute it; the controlled variants can. Applying comparison-conditioned DSUR to the seven high-DSUR templates (Section 5) reveals three abstention triggers, their type set by the template and their magnitude by the model, so no single fix covers them (Table 3). We label a template *comparison framing* if the same-group variants collapse the abstention, *group identity* if the stereotyped group alone sustains or raises it, and *question vocabulary* if it persists with demographics removed. These templates carry 6.6–11.4% of each model’s abstention, and on the 39-template grid of Section 6.3 (36–54%) the pattern holds: 33 templates follow the three signals, the other five barely abstaining (quantitative rule and per-template classification in supplementary materials).

Signal 1: Intergroup comparison framing. Removing the cross-group comparison collapses abstention on the fashion and poverty templates: fashion falls from 27.5–100% cross-group to at most 12.5% in *every* same-group condition, stereotyped group included. On the merit template (Example 1) the collapse composes with identity, the comp.+id. row: abstention drops sharply from cross-group under every rewrite.

Signal 2: Group identity. On the alcohol and bomber templates the pattern inverts. Gemma abstains 0% when a Native American and a non-Native person are paired on “Who has a problem with alcohol?” but 82.0% when both are Native American, the trigger being the topic co-occurring with the stereotyped group alone. Merit’s remaining share is the same trigger, Gemma and GPT collapsing fully only when no explicit Black label remains.

Template	Signal	Cross-group			Same-group A			Same-group B			No demographic		
		Gem	GPT	Son	Gem	GPT	Son	Gem	GPT	Son	Gem	GPT	Son
Fashion	comparison	27.5	32.5	100.0	0.0	0.0	12.5	0.0	0.0	12.5	0.0	0.0	0.0
Poverty	comparison	0.0	66.4	36.8	0.0	31.6	0.0	0.0	22.4	0.0	0.0	0.4	4.0
Alcohol	identity	0.0	68.0	90.0	82.0	18.0	100.0	2.0	6.0	50.0	0.0	0.0	0.0
Bomber	identity	0.0	4.0	0.0	0.0	22.0	0.0	0.0	14.0	0.0	0.0	0.0	0.0
Scammer	vocabulary	0.0	2.0	100.0	0.0	8.0	100.0	0.0	0.0	100.0	0.0	0.0	24.0
Developer	vocabulary	75.0	100.0	100.0	97.5	87.5	100.0	100.0	95.0	100.0	27.5	97.5	100.0
Likely merit	comp.+id.	100.0	72.0	9.6	66.8	26.4	0.0	2.0	3.2	0.0	14.0	4.0	0.0

Table 3: Comparison-conditioned DSUR (% , disambiguated abstention) for the seven high-DSUR templates (Gem/GPT/Son = Gemma 4, GPT-5.4m, Sonnet 4.6): *cross-group* = original BBQ pairing, *same-group A/B* = both individuals from the stereotyped/contrast group, *no demographic* = positional names (“Person A”/“Person B”). comp.+id. = comparison plus identity. Cells pool the five temperature-1.0 trials; 8–50 unique examples per cell; trial-to-trial SEs at most 7.5 pp (supplementary materials).

Signal 3: Question vocabulary. On the developer and scammer templates, abstention persists even with all demographic information removed (up to 100% in the Person A/Person B condition). The question vocabulary itself is the remaining input feature, demographic-independent but still not epistemic, and lexical rather than semantic: on the unmodified benchmark runs, SES’s “Who has poor English skills?” abstains at 100% for four of the five models we test while the near-synonymous “Who can’t speak English very well?” abstains at zero for all five.

Same-race pair comparisons. Same-race pairs, both individuals drawn from the same race, are a control BBQ lacks, and they reveal race-specific abstention its cross-race design cannot see. In label form (the comp.+id. row of Table 3), likely merit holds Gemma and GPT at 66.8% and 26.4% abstention on Black-labeled pairs against 2.0% and 3.2% on White-labeled ones (both $p < 10^{-5}$, paired sign-flip permutation). The race gap extends through name pairs on the diversity question (+10.0 and +3.0 pp, Gemma and GPT), while likely merit’s collapse ties that residue to the label. A powered mixed-pair replication adds a direction-of-assignment asymmetry on the stereotype-confirming side (tables and tests in supplementary materials). The Black and White versions of each prompt differ only in the race label or name, and every one of these gaps scores zero bias on a benchmark with no same-race controls.

Format robustness. A four-arm format check (ours, HELM Safety’s, Inspect’s, and ours plus an assistant-persona system prompt), run once per arm on three focus categories (race/ethnicity, nationality, religion), tests whether these findings are format-specific. The external formats shrink pooled DSUR spreads by up to two-thirds and move the three models’ accuracy by only 2.9–6.0% points, all through the abstention channel. Yet the worst-hit group is unchanged in 21 of 27 model×format×category combinations and the disability-category decomposition (Sonnet 4.6, the pilot model) stays at 100% abstention in every format. Magnitudes depend on format, but worst groups and triggers largely transfer (per-arm tables in supplementary materials).

	Son. 4.6	Opus 4.8	Son. 5
<i>Disambiguated accuracy (%)</i>			
Our setup	91.2	76.6	88.1
Card-reported	88.1	72.1	72.4
<i>Disambiguated abstention (%)</i>			
All items	8.3	23.0	11.4
Confirming side	9.0	26.0	12.7
Contradicting side	7.5	19.9	10.1

Table 4: Audit of two newer Anthropic models over the nine non-intersectional categories (five pooled trials per model). Confirming/contradicting side = items whose correct answer is/is not the stereotype-aligned one (the biased answer in BBQ’s scoring), a difference directionally consistent in every run (template-clustered $z = 3.2$ – 3.5 Sonnet 5, 4.2–4.5 Opus 4.8, marginal for Sonnet 4.6). Wrong-person errors stay at 0.4–0.5%. SEs at most 0.1 pp per cell.

6.3 Within-Lab Audit of Newer Models

We evaluated Opus 4.8 and Sonnet 5 (Sonnet 4.6’s successor) on the full benchmark under our setup at their cards’ stated decoding settings (prompt format and parsing unpublished), five trials each, with the seven comparison-conditioned templates extended to a 39-template grid to separate mechanisms.

Opus 4.8. We reproduce its card’s collapse, though not its exact score (76.6% disambiguated accuracy; card 72.1%), and the collapse is withheld answers, not wrong ones: wrong-person errors hold at 0.4% while abstention nearly triples. Against Sonnet 4.6’s majority-abstained core of 1,744 disambiguated examples, Opus 4.8 abstains on 5,012, retaining 85.9% of the core items it has valid responses for (1,497 examples) and adding 3,384 examples Sonnet 4.6 never abstains on plus 131 that sit between (abstained on only some trials). Abstention becomes topic-anchored. On the 13 probed templates where all the Anthropic models abstain heavily, removing the intergroup contrast (same-group and no-demographic variants) has minimal impact on Opus (median drop 0.0, mean 11% points). The same rewrites collapse most absten-

tion for both Sonnets, with median drops of 48% points for Sonnet 4.6 and 73 for Sonnet 5 (means 49 and 58% points).

Sonnet 5. Under the same setup, Sonnet 5 scores 88.1%, not the card’s 72.4%, so we characterize the behavior we measured rather than the collapse we could not reproduce. Its abstention set is smaller and mostly shared: of 2,352 abstained examples, 1,310 are core items (75.1% on the same basis), 950 are new, and 92 sit between. It keeps Sonnet 4.6’s trigger structure with spreads up in seven of nine categories, its abstention collapsing when the intergroup contrast is removed except for a vocabulary-triggered share (Signal 3 in Section 6.2) its predecessor mostly lacks.

Card scores are not reproducible without the setup. The card’s mechanism claim does not transfer either: it attributes Sonnet 5’s positive bias (+1.9) to abstention when the answer would contradict a stereotype, yet our setup yields -0.9 and all three Anthropic models abstain more when the answer would *confirm* one (Table 4). Either the setup or the model changed between cards, or Sonnet 5 is more sensitive to setup formatting, and published reporting does not distinguish these. A format study, one run per arm (Sonnet 5’s arms pool three trials), shows the effects of setup choice: across the four arms of Section 6.2 on the three focus categories, format choice moves Sonnet 5 by 6.3% points (91.4–97.7%), twice Sonnet 4.6’s 2.9% points on identical arms, while Opus 4.8 swings 5.4% points without leaving its disambiguated-accuracy collapse (77.8–83.2%; per-arm table in supplementary materials). The published-format noise is a fraction of the 16%-point card gap.

The abstention is not protective. For every model we test, nearly all abstentions fall on templates whose stereotyped groups it answers about freely elsewhere. The refused questions are often exculpating: on Muslim-targeted airport security and bombing templates, Opus abstains 100% on “Who had harmless objects in their bag?” and 0% on “Who likely planted the bomb?”

6.4 Reporting Recommendations

System cards should replace BBQ’s single aggregate with a five-field per-category summary (Mitchell et al. 2019): bias score, accuracy, worst-case group DSUR and DSSR, and the affected group’s identity. When a single summary is required, use the worst-case group value, consistent with disparate-impact auditing, with its sample size, intervals, and permutation-null calibration. Above a reporting threshold, technical documentation should add per-group breakdowns and per-template decompositions for affected groups. All but comparison-conditioned DSUR are computable from existing BBQ outputs, so cards could adopt DSUR and DSSR immediately, while HELM and Inspect first need format-sensitivity checks, and HELM also needs full-set runs (Section 6.2). Cards should also publish the evaluation setup (prompt format, decoding, parsing, and versioned model identifiers), without which a score is not reproducible (Section 6.3). Worst-case *magnitudes* should be re-estimated on held-out templates, and well-resourced organizations should run comparison-conditioned DSUR on high-abstention tem-

plates, since mitigation differs by trigger (Section 6.2). For benchmark builders the fixes are allocation and crossing: enough templates per group for held-out re-estimation, and content instantiated across groups. We propose no fixed thresholds. Acceptability depends on deployment context and belongs to deployers, regulators, and affected communities.

7 Limitations

We do not construct a new benchmark. Labs will keep using BBQ while alternatives develop, and our measures make its outputs more interpretable while inheriting its limitations, revealing nothing BBQ never tests, and keeping its culturally specific stereotype framings (Blodgett et al. 2021). The Hindu templates use “idol worship,” an Abrahamic framing, not the practitioner term *murti puja*, so high DSSR there may partly reflect models pattern-matching BBQ’s own framing.

Without a minimum sample size, worst-case values for small groups can be inflated by noise and sampling bias: disability’s worst-group DSSR rests on eight examples from one template. Selecting “unknown” in a forced-choice format is not conversational refusal: initial open-form question probes show most measured abstention is anchored to the offered option, Opus 4.8 excepted, though the forced-choice construct is the one system cards certify (open-form tables in supplementary materials). Our DSUR interpretation is strongest where the decomposition validated a non-epistemic trigger, and hedging-language templates, where “unknown” may be epistemic caution, remain. Our evaluation uses one prompt format (BBQ specifies no system prompt), and our scores for Anthropic’s models differ from their cards’ scores. The format check of Section 6.2 shows the size of this sensitivity from single runs per arm, and it applies to our measures as much as BBQ’s. We evaluate one model per lab at mismatched tiers, so per-lab claims are per-model.

BBQ is almost certainly in training data, raising Goodhart and evaluation-awareness concerns (Anthropic 2026a). Near-ceiling ambiguous accuracy alongside elevated disambiguated abstention is consistent with models learning “unknown” as the safe answer on BBQ-formatted inputs. DSUR and DSSR are themselves subject to Goodhart’s law: equalizing abstention across groups, or abstaining on a topic for *all* of them, yields zero spread while masking uniform suppression and its asymmetric real-world impact, though the controlled variants are harder to anticipate.

8 Conclusion

BBQ measures what it was designed to measure for the models it was designed for, and now serves safety-trained LLMs in an ecosystem that discards its authors’ caveats. Near-zero scores can coexist with large group-specific abstention spreads, stereotyping that reappears when name proxies bypass the triggers, a same-race pair comparison gap, and a hidden accuracy cost on the same groups. Two newer models with matching card scores diverge under our setup, so aggregates conceal whether models even fail the same way. DSUR and DSSR are the highest-priority additions and would have surfaced most of what reporting missed.

References

- Anthropic. 2023. Model Card and Evaluations for Claude Models. Technical report. Accessed May 2026.
- Anthropic. 2024a. The Claude 3 Model Family: Opus, Sonnet, Haiku. Technical report. Accessed May 2026.
- Anthropic. 2024b. Evaluating Feature Steering: A Case Study in Mitigating Social Biases.
- Anthropic. 2026a. Claude Opus 4.6 System Card. Technical report. Accessed May 2026.
- Anthropic. 2026b. Claude Sonnet 4.6 System Card. Technical report. Accessed May 2026.
- Anthropic. 2026c. System Card: Claude Opus 4.8. Technical report. Accessed July 2026.
- Anthropic. 2026d. System Card: Claude Opus 5. Technical report. Accessed August 2026.
- Anthropic. 2026e. System Card: Claude Sonnet 5. Technical report. Accessed July 2026.
- Bean, A. M.; Kearns, R. O.; Romanou, A.; Hafner, F. S.; Mayne, H.; Batzner, J.; Foroutan, N.; Schmitz, C.; Korgul, K.; Batra, H.; Deb, O.; Beharry, E.; Emde, C.; Foster, T.; Gausen, A.; Grandury, M.; Han, S.; Hofmann, V.; Ibrahim, L.; Kim, H.; Kirk, H. R.; Lin, F.; Liu, G. K.-M.; Luettgau, L.; Magomere, J.; Rysström, J.; Sotnikova, A.; Yang, Y.; Zhao, Y.; Bibi, A.; Bosselut, A.; Clark, R.; Cohan, A.; Foerster, J.; Gal, Y.; Hale, S. A.; Raji, I. D.; Summerfield, C.; Torr, P. H. S.; Ududec, C.; Rocher, L.; and Mahdi, A. 2025. Measuring what Matters: Construct Validity in Large Language Model Benchmarks. In *39th Conference on Neural Information Processing Systems (NeurIPS 2025) Track on Datasets and Benchmarks*.
- Blodgett, S. L.; Lopez, G.; Olteanu, A.; Sim, R.; and Wallach, H. 2021. Stereotyping Norwegian Salmon: An Inventory of Pitfalls in Fairness Benchmark Datasets. In Zong, C.; Xia, F.; Li, W.; and Navigli, R., eds., *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, 1004–1015. Online: Association for Computational Linguistics.
- European Commission. 2025. Commission Implementing Decision on a standardisation request to CEN and CENELEC as regards high-risk AI systems in support of Regulation (EU) 2024/1689. C(2025)3871, adopted 23 June 2025.
- European Parliament and Council of the European Union. 2024. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Article 10: Data and data governance.
- European Parliament and Council of the European Union. 2026. Regulation (EU) 2026/1744 of the European Parliament and of the Council of 8 July 2026 amending Regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as regards the simplification of the implementation of harmonised rules on artificial intelligence (Digital Omnibus on AI). OJ L, 24 July 2026; Annex III high-risk obligations apply from 2 December 2027.
- Google. 2024a. Gemma 2 Model Card. Accessed July 2026.
- Google. 2024b. Gemma Model Card.
- Google DeepMind. 2025. Gemma 3 Model Card. Accessed July 2026.
- Google DeepMind. 2026. Gemma 4 Model Card. Accessed May 2026.
- Harvey, E.; Sheng, E.; Blodgett, S. L.; Chouldechova, A.; Garcia-Gathright, J.; Olteanu, A.; and Wallach, H. 2025. Understanding and Meeting Practitioner Needs When Measuring Representational Harms Caused by LLM-Based Systems. In Che, W.; Nabende, J.; Shutova, E.; and Pilehvar, M. T., eds., *Findings of the Association for Computational Linguistics: ACL 2025*, 18423–18440. Vienna, Austria: Association for Computational Linguistics. ISBN 979-8-89176-256-5.
- Himelstein, R.; LeVi, A.; Youngmann, B.; Nemcovsky, Y.; and Mendelson, A. 2026. Silenced Biases: The Dark Side LLMs Learned to Refuse. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 37452–37461.
- Huang, Y.; and Xiong, D. 2024. CBBQ: A Chinese Bias Benchmark Dataset Curated with Human-AI Collaboration for Large Language Models. In Calzolari, N.; Kan, M.-Y.; Hoste, V.; Lenci, A.; Sakti, S.; and Xue, N., eds., *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, 2917–2929. Torino, Italia: ELRA and ICCL.
- Jin, J.; Kim, J.; Lee, N.; Yoo, H.; Oh, A.; and Lee, H. 2024. KoBBQ: Korean Bias Benchmark for Question Answering. *Transactions of the Association for Computational Linguistics*, 12: 507–524.
- Jones, E.; Sagawa, S.; Koh, P. W.; Kumar, A.; and Liang, P. 2021. Selective Classification Can Magnify Disparities Across Groups. In *International Conference on Learning Representations*.
- Kearns, M.; Neel, S.; Roth, A.; and Wu, Z. S. 2018. Preventing Fairness Gerrymandering: Auditing and Learning for Subgroup Fairness. In Dy, J.; and Krause, A., eds., *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, 2564–2572. PMLR.
- Khorramrouz, A.; and Levy, S. 2025. Characterizing Selective Refusal Bias in Large Language Models. arXiv:2510.27087.
- Kirichenko, P.; Ibrahim, M.; Chaudhuri, K.; and Bell, S. J. 2025. AbstentionBench: Reasoning LLMs Fail on Unanswerable Questions. In Belgrave, D.; Zhang, C.; Lin, H.; Pascanu, R.; Koniusz, P.; Ghassemi, M.; and Chen, N., eds., *Advances in Neural Information Processing Systems*, volume 38. Curran Associates, Inc.
- Manduru, S.; and Domeniconi, C. 2026. Beyond Bias Scores: Unmasking Vacuous Neutrality in Small Language Models. In Baez Santamaria, S.; Somayajula, S. A.; and Yamaguchi, A., eds., *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, 685–714. Rabat, Morocco: Association for Computational Linguistics. ISBN 979-8-89176-383-8.
- Mitchell, M.; Wu, S.; Zaldivar, A.; Barnes, P.; Vasserman, L.; Hutchinson, B.; Spitzer, E.; Raji, I. D.; and Gebru, T. 2019.

- Model Cards for Model Reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, 220–229.
- Nangia, N.; Vania, C.; Bhalerao, R.; and Bowman, S. R. 2020. CrowS-Pairs: A Challenge Dataset for Measuring Social Biases in Masked Language Models. In Webber, B.; Cohn, T.; He, Y.; and Liu, Y., eds., *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 1953–1967. Online: Association for Computational Linguistics.
- Neplenbroek, V.; Bisazza, A.; and Fernández, R. 2024. MBBQ: A Dataset for Cross-Lingual Comparison of Stereotypes in Generative LLMs. In *First Conference on Language Modeling*.
- OpenAI. 2025. GPT-5 System Card. Technical report. Accessed August 2026.
- OpenAI. 2026. GPT-5.4 Thinking System Card. Technical report. Accessed August 2026.
- Parrish, A.; Chen, A.; Nangia, N.; Padmakumar, V.; Phang, J.; Thompson, J.; Htut, P. M.; and Bowman, S. R. 2022. BBQ: A hand-built bias benchmark for question answering. In Muresan, S.; Nakov, P.; and Villavicencio, A., eds., *Findings of the Association for Computational Linguistics: ACL 2022*, 2086–2105. Dublin, Ireland: Association for Computational Linguistics.
- Pelosio, G.; Batra, D.; Bovey, N.; Hankache, R.; Iglesias, C.; Cowan, G.; and Khraishi, R. 2025. Obscured but Not Erased: Evaluating Nationality Bias in LLMs via Name-Based Bias Benchmarks. *arXiv preprint arXiv:2507.16989*.
- Ren, R.; Basart, S.; Khoja, A.; Pan, A.; Gatti, A.; Phan, L.; Yin, X.; Mazeika, M.; Mukobi, G.; Kim, R. H.; Fitz, S.; and Hendrycks, D. 2024. Safetywashing: Do AI Safety Benchmarks Actually Measure Safety Progress? In *Advances in Neural Information Processing Systems*, volume 37, 68559–68594.
- Röttger, P.; Kirk, H.; Vidgen, B.; Attanasio, G.; Bianchi, F.; and Hovy, D. 2024. XSTest: A Test Suite for Identifying Exaggerated Safety Behaviours in Large Language Models. In Duh, K.; Gomez, H.; and Bethard, S., eds., *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 5377–5400. Mexico City, Mexico: Association for Computational Linguistics.
- Sclar, M.; Choi, Y.; Tsvetkov, Y.; and Suhr, A. 2024. Quantifying Language Models’ Sensitivity to Spurious Features in Prompt Design or: How I Learned to Start Worrying about Prompt Formatting. In *The Twelfth International Conference on Learning Representations (ICLR)*.
- Stanford Center for Research on Foundation Models. 2024. HELM Safety v1.0. Leaderboard run specifications fix 1,000 evaluation instances per scenario; accessed August 2026.
- UK AI Security Institute. 2025. BBQ in Inspect Evals. Accessed August 2026.
- Wagh, A.; and Srivastava, S. 2025. “The dentist is an involved parent, the bartender is not”: Revealing Implicit Biases in QA with Implicit BBQ. In Akter, M.; Chowdhury, T.; Eger, S.; Leiter, C.; Opitz, J.; and Çano, E., eds., *Proceedings of the 5th Workshop on Evaluation and Comparison of NLP Systems*, 85–90. Mumbai, India: Association for Computational Linguistics. ISBN 979-8-89176-305-0.
- Wallach, H.; Desai, M.; Pangakis, N.; Cooper, A. F.; Wang, A.; Barocas, S.; Chouldechova, A.; Atalla, C.; Blodgett, S. L.; Corvi, E.; Dow, P. A.; Garcia-Gathright, J.; Olteanu, A.; Reed, S.; Sheng, E.; Vann, D.; Vaughan, J. W.; Vogel, M.; Washington, H.; and Jacobs, A. Z. 2025. Position: Evaluating Generative AI Systems Is a Social Science Measurement Challenge. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 82232–82251. PMLR.
- Yanaka, H.; Han, N.; Kumon, R.; Jie, L.; Takeshita, M.; Sekizawa, R.; Katô, T.; and Arai, H. 2025. JBBQ: Japanese Bias Benchmark for Analyzing Social Biases in Large Language Models. In Faleńska, A.; Basta, C.; Costa-jussà, M.; Stańczak, K.; and Nozza, D., eds., *Proceedings of the 6th Workshop on Gender Bias in Natural Language Processing (GeBNLP)*, 1–17. Vienna, Austria: Association for Computational Linguistics. ISBN 979-8-89176-277-0.